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Calibra: A Robotic System for Automated AnalysisClark Zhang,^{*} Matthew Reish and Jason E. Hein

Received 00th January 20xx,

Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

We have designed and built Calibra

1 Introduction

Automation and robotics are rapidly changing the field of analytical chemistry, and the benefits of automation are readily apparent. Automated sample preparation can address many challenges associated with manual procedures as they are capable of providing consistent, reproducible results that reduce human error.^{1–3} By relegating tedious manual tasks to automated systems, researchers are able to concentrate on more intellectually demanding tasks and can reduce exposure to potentially hazardous substances.^{4–6}

Automated sample preparation with online analysis is currently the exception rather than the norm, particularly for the analysis of small molecules and pharmaceuticals which lag significantly behind their bioanalytical counterparts.⁷ A survey of analytical labs suggests that at least 60% of time spent on analysis is taken up by sample preparation and, perhaps more importantly, at least 50% of error in analytical analysis come from human error during sample preparation.^{8,9} This suggests that the time and resources spent on the automation of sample preparation would be well spent. Several commercial suppliers have products available that can effectively perform end-to-end analysis including sample preparation and injection onto analytical instruments. However, these platforms are typically custom built for each workflow, making them prohibitively expensive and specialized for a single type of analysis.¹⁰

The increasing accessibility of open-source software and hardware has played a significant role in accelerating the innovation of robotic automated analytical chemistry research.¹¹ Advances such as 3D printing and easily accessible, low-cost microcontrollers have catalyzed the "fourth industrial revolution", enabling researchers to build and use modules with minimal training.¹²

The open-source community currently offers a variety of laboratory preparatory modules including liquid handlers, vortex mixers, and centrifuges. In the case of liquid handlers, only a few examples of liquid handling on non-cartesian robotic arms have been described.¹³ Similarly, while other preparatory equipment, like vortex mixers and centrifuges exist in the open-source community, these examples are limited in user and positional control, often requiring manual intervention between operations.^{14–16} As such, while open-source modules are widely accessible, they often cause bottlenecks due to the lack end-to-end integration.^{17,18} This limitation poses one of the most immediate challenges to the massive deployment of automated laboratories, where dynamically integrated components are critical.¹⁹ Developing adaptable, modular, and cost-effective components that address the rigidity in the fixed sampling elements are therefore critical to advance automation in small molecule analytical workflows.²⁰

As a step towards more robust, modular, and open-source automated platforms in small molecule chemistry, we developed Calibra— an automated system designed to perform end-to-end sample preparation and accurately determine the weight percentage of an unknown sample within a library, requiring minimal user input. Since this procedure is traditionally performed manually, we adopted a human-inspired approach to inform the designs for the module designs. By automating all the time-consuming sample preparation steps,

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Supplementary Information available: [details of any supplementary information available should be included here]. See DOI: 10.1039/x0xx00000x

the process requires user input only for adding an arbitrary quantity of the compound of interest to a pre-tared vial, after which the system completes the analysis. Expansion of the compound library is equally straightforward, as calibration curves are automatically generated by providing the robot with a sample of known concentration and its corresponding retention time.

Calibra, apart from offering an end-to-end solution for calibration curve generation and quantification, was developed from custom-designed automated modules for centrifuging, vortex mixing, liquid handling, and sample tracking to minimize repetitive manual sample preparation steps, saving researchers valuable time and increasing processing efficiency.^{21,22} The platform leverages open-source, modular components, all integrated through custom, open-source Python software. The software also incorporates commercial components, like precision weighing instruments and robotic vial transport arms, programmed such that each of our elements can be implemented into other workflows with no need for redesigns or modifications. After validating the system's reliability, we further applied it to quantify real-world community drug samples, demonstrating its practical utility and robustness. By prioritizing accessibility and modularity, Calibra supports the democratization of scientific research, making advanced analytical tools available to a broader audience while maintaining cost-effectiveness and versatility.

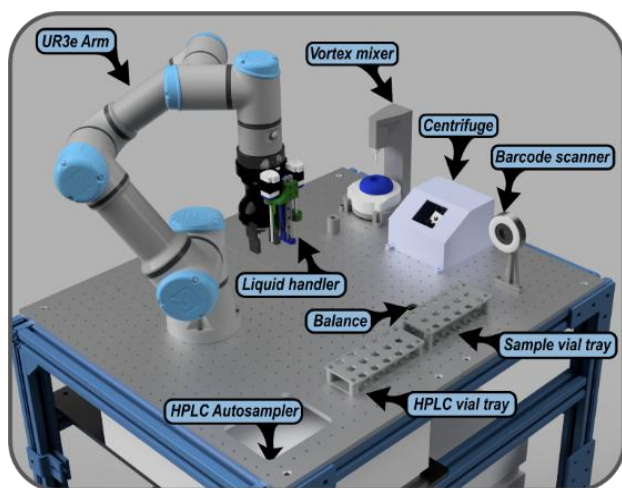


Figure X: Render of the robotic system with modular elements described in the text labelled.

2.1 Modular Hardware Elements

To recreate the manual workflows described above, six modular elements are used to precisely prepare samples for analytical analysis.

2.1.1 Liquid Handler To the best of our knowledge, the liquid handler described here represents the first robotic arm mounted system to have a retractable platform to minimize interference with the grippers, as well as a retractable needle, facilitating septa piercing without unwanted vial movements. Figure XA shows the retraction mechanism which includes two stepper motor linear actuators to extend the platform and the needle separately. The needle is connected by 0.04-inch ID PTFE tubing to a Cavro Xcalibur syringe pump with a 1 mL syringe.²³

The stepper motors are controlled by a custom motor driver developed in-house which includes two Trinamic TMC2226 drivers for motor control and an Espressif ESP-32 module for communication and control of the stepper drivers. The firmware for the driver control board was also written in-house. The firmware interprets serial commands sent from python and translates them directly into stepper movements. Both the hardware design and firmware are freely available and a full description can be found in the SI.²⁴

2.1.2 Vortex mixer Our system required an automated module to ensure dissolution and sample homogeneity. A common method for dissolution in other self-driving labs include commercial sonicators, operated either manually,²⁵ or automatically by the robot.²⁶ In previous testing, we found that variations in liquid level in the sonicator bath during periods of extensive autonomous operation could lead to loss of reliable sample mixing, which prompted us to opt for a vortex mixer, ideal in situations of dissolution and mixing at small scales.²⁷

A commercially available Fisherbrand Mini Vortex Mixer (model number BV202) was adapted with a 3D-printed mount which uses Nanotec linear actuator (model number LGA201S06-A-TDBA-019) to first lightly hold the sample vial in place after the vial is delivered by the robot arm. Following delivery of the vial and withdrawal of the robotic arm, the actuator then pushes the vial further to engage the vortexer's pressure switch to initiate vortex mixing. The linear actuator is controlled by a custom-designed stepper motor driver which is discussed above in section xxxx. The design of the vortexer is detailed in figure XB.

2.1.3 Centrifuge The separation of insoluble impurities is handled by a centrifuge. While alternatives are used in other systems like gravity filtration²⁸ and vacuum filtration,²⁹ centrifuge systems have comparatively higher separation performance, time efficiency, and space efficiency.³⁰

2 Methods

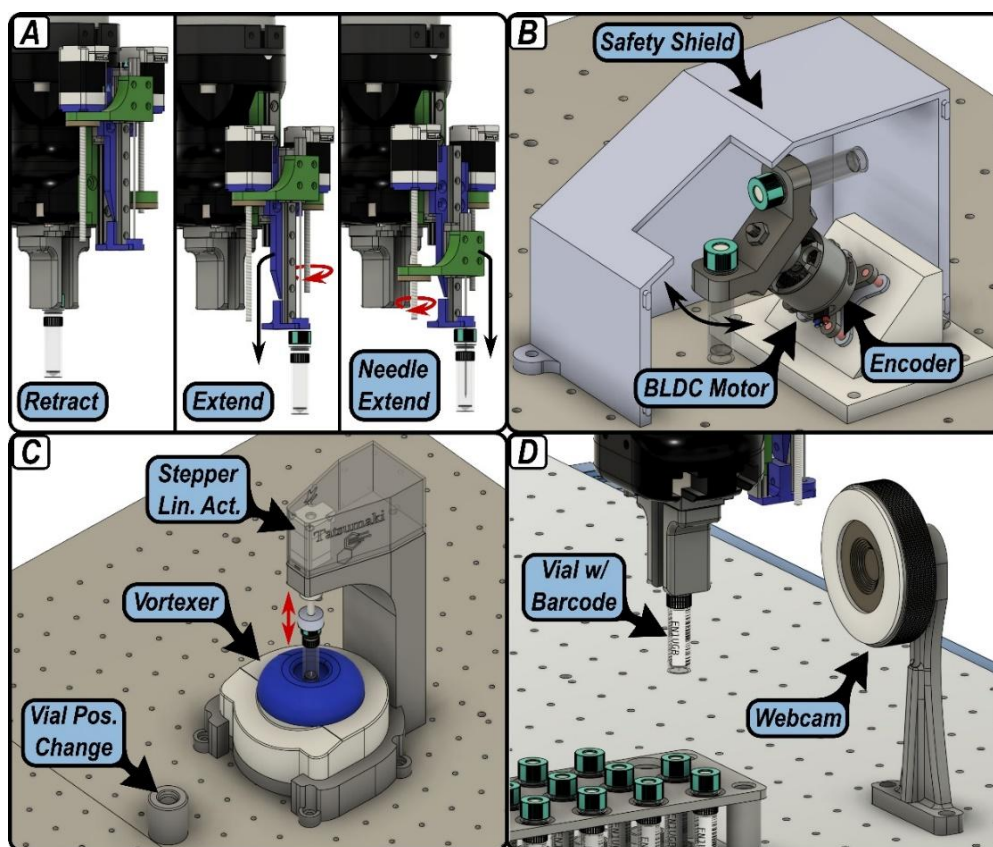


Figure X Details of the operation of each module. A: Shows the operation of the liquid handler. A: Left: The retract position is shown which gives access to the grippers. A: Middle: The stepper motor linear actuator on the right extends the base of the system, which rides on a linear rail. In the extend position the needle foot is lower than the tip of the gripper fingers. A: Right: The stepper motor linear actuator on the left extends the needle, which rides on a second linear rail, to pierce the septum of a vial. B: Detailed view of the centrifuge with the half of the safety shield removed. The position of the magnetic encoder is highlighted. The encoder allows the rotor to return to the same position after each run, allowing the arm to retrieve and place vials. C: The automated vortexer is shown highlighting the operation of the linear actuator which holds the vial and activates the vortexer. D: The elements of the barcode sample tracking system are shown including a webcam and a barcode-labelled vial.

Current commercial benchtop centrifuges often present limitations that make them unsuitable for automation applications.¹⁴ Centrifuges suitable for automation must return to a pre-set position after a run and be operable by remote commands. While several commercially available models meet these criteria, they are generally costly and space inefficient. In response to these shortcomings, we developed a custom centrifuge that is simple to set up, and offers a high level of user control, including the ability to adjust speed, acceleration, number of rotations, and advanced proportional-integral-derivative PID control for motor finetuning. This system also features essential functionalities like speed tracking and automated positioning. The design of the centrifuge is shown in Figure XB.

The centrifuge uses a commercially available SunnySky X2212 brushless direct-current (BLDC) drone motor that spins at

speeds up to 12000 rpm, achieving a relative centrifugal force of 1610 g-forces. The motor is fitted with a TLE 5012 magnetic encoder with an output of 4096 counts per revolution, which allows determination of its angular rotation at any given time and return to the home position without additional input. The system is controlled by an O-Drive BLDC motor controller clone and powered by a 24V power supply.^{31,32}

For safety, a custom 3D-printed shell was designed to enclose the centrifuge. In case of system failure during high-speed operation, the enclosure minimizes the chances of injury. The housing includes a strategically placed opening that enables robot access during vial transfers.

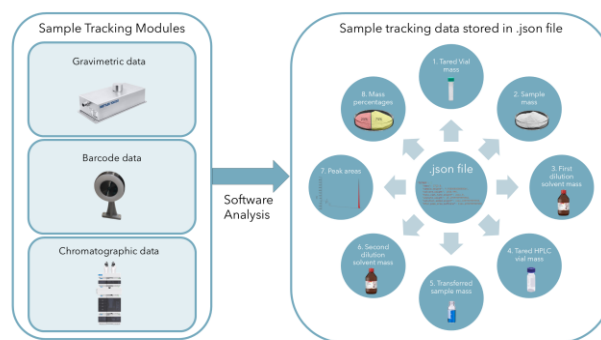
2.1.4 Balance A Mettler-Toledo WKC204C weighing module with 0.1 mg readability is used as part of the larger sample tracking system to tare vials and measure the mass of the sample before preparation. Additionally, between each solvent delivery step, the balance is used to record the mass of

delivered solvent with accuracies on the order of 0.1 μl . This is at least an order of magnitude better than the accuracy performance expected from a syringe pump. To save space, the balance was mounted on custom-designed legs on the underside of the aluminum platform, with a hole drilled through the platform to allow the robot to insert vials directly onto the weighing surface.

2.1.5 Sample tracking system Interlab and intralab sample tracking has previously been accomplished using QR codes, as well as barcodes.^{33–35} While QR codes offer a higher number of unique combinations, initial testing showed difficulties scanning QR codes when they are placed on the curved surface of a vial. Our system used a Python script to generate random barcodes that are six characters long. The barcodes were printed on Brother DK2212 film tape. This worked well for printed barcodes as it adhered well to curvature of vials. After testing, the tared vials were found to not deviate significantly from the tared mass even after long terms of storage or handling, at least to the 0.1 mg resolution achieved by the system.

Our barcode scanner uses a commercially available Razer Kiyo webcam (model number RZ19-02320) attached to a 3D printed mount. At the beginning of each run, the robot presents the vial to the camera and the vial's barcode is scanned. The data from the balance and HPLC are recorded in a JSON file under the previously scanned barcode for later analysis. After completion of HPLC analysis, peak area data is extracted and appended to the corresponding barcode entry for use in calibration curve and weight percentage calculations. For calibration curve generation, the peak areas are plotted against the known concentrations to create the curve. In quantification assays, the established calibration curve is then used to calculate the weight percentage of the analyte based on its peak area. The structure of the JSON file, in the order of the workflow, can be seen in Figure X.

2.1.6 Robot A six degree-of-freedom (6DoF) robotic arm replaces the human operator to handle vial transfers between vial trays, balance, barcode scanner, vortexer, and centrifuge. Dilutions and solvent transfer are handled by a liquid handling module that is attached to the robotic arm, again mimicking solvent delivery by a human operator. For the robot arm we selected a Universal Robots UR3e robotic arm equipped RobotIQ Hand-E grippers for its ability for high-precision movement and flexibility, making it ideal for repetitive, delicate tasks such as vial handling. However, the robotic arm is interchangeable based on budget and project requirements. For projects with more limited resources, alternative robots of vial transport like could be used, while still maintaining the core functionality of the workflow constituted by the modular elements.



2.2 Software

The main workflows are entirely handled by the top-level Python functions *make_calibration_curve* for calibration curve generation and *quant_solid* for solid quantification. Variables in the function allows users to have a high level of control. It is possible to specify the volume used to dilute the centrifuge and HPLC vials, volume transferred between the centrifuge and HPLC vials, number and position of vials analyzed, and type of compound being analyzed, if applicable. All the code can be found on Hein Lab GitLab.³⁶ Some important software modules in our code are described below.

2.2.1 Robot control The UR3e robot code uses a Python library developed by Hein Lab.³⁷ The library handles robot linear and joint movements to manually-defined locations stored in a script file. For further customization, gripper, speed, and acceleration control are also built into the library.

2.2.2 Software Interfacing A common problem in automation is the integration of hardware with software that require manual input. In the case of Agilent OpenLab CDS used in our system, we opted to use the open-source PyAutoGUI library to automate the operation of the HPLC system.³⁸ PyAutoGUI was employed to launch the HPLC acquisition software, control the autosampler injector, initiate runs, and check for run completion. Predefined screenshots of the necessary interface buttons are stored and used in our code. However, during testing, we identified that PyAutoGUI was a potential source of failure, as it always needs to be in the foreground to function. As a result, we recommend that users that use OpenLab Chemstation instead of OpenLab CDS in order adopt the custom macro for more reliable operation. Our macros are developed partly from a module from the Chemputer system developed by Gromski et al.³⁹

2.2.3 Graphical User Interface For operational simplicity, our system uses ivoryOS, a user-friendly, web-based graphical user interface (GUI) to simplify our human-robot interaction.⁴⁰ In ivoryOS, function cards mirror the Python code where users can easily enter specific variable values into text boxes and click a single button to initiate the entire workflow. ivoryOS offers a streamlined solution for reducing the learning curve of new

users and eliminates the need to understand every Python function to use our modules.

2.2.4 Centrifuge The centrifuge code uses the public ODrive Python library to detect and operate the motor.³² Our code supports velocity and acceleration control, as well as a built in proportional-integral-derivative (PID) algorithm for positional error correction. We recommend that users of our code tune the PID control to avoid motor gyration to prolong system lifespan.⁴¹

2.2.5 Data Processing Our custom-built calibration curve plotting and quantification algorithms operate using data files generated by Agilent software processing methods. Once the compound of interest is identified in the processing method, the Agilent software can output a CSV file after each HPLC run that includes the compound name along with the corresponding peak area (assuming the compound was part of the analyte). Our algorithms therefore take advantage of this feature to automatically search for and extract this peak area for use in both calibration curve generation and subsequent quantification workflows.

For calibration curve generation, it is necessary that the user identify the compound of interest both in the online GUI and in the HPLC processing method. When all of the HPLC runs finishes, our code scans all CSV files for each run and locates peak areas associated with the specified compound name. The values are stored as the Y-axis data points of our calibration curve. Simultaneously, the concentration values are calculated using values taken from the sample tracking JSON data file, as shown in section 2.1.5. The calculated concentration values provide the X-axis values. The data is then automatically plotted to generate the calibration curve, and the resulting slope or response factor is stored in a separate JSON file for later quantification.

The quantification algorithm follows a similar procedure but does not require users to specify a compound name. Instead, the algorithm cross-references the compound names in the CSV files generated by the HPLC run with those in the JSON database previously described and extracts the relevant peak areas. Using the calibration curve gradient stored in the JSON database, the algorithm converts the peak areas into concentrations. Each concentration is then processed to determine mass percentage by multiplying by the solvent volume and dividing by the total mass.

2.3. Autonomous quantification assays and calibration curve generation

The hardware and software elements described in section 2.1 and 2.2 respectively are integrated into our autonomous workflow. The workflow is designed to mimic the manual procedure, with additional gravimetric determination steps to correct for any possible deviations. First, as part of the manual preparation step, the vial is labelled with a barcode and tared,

storing the tared weight in a JSON file. Then, the user must

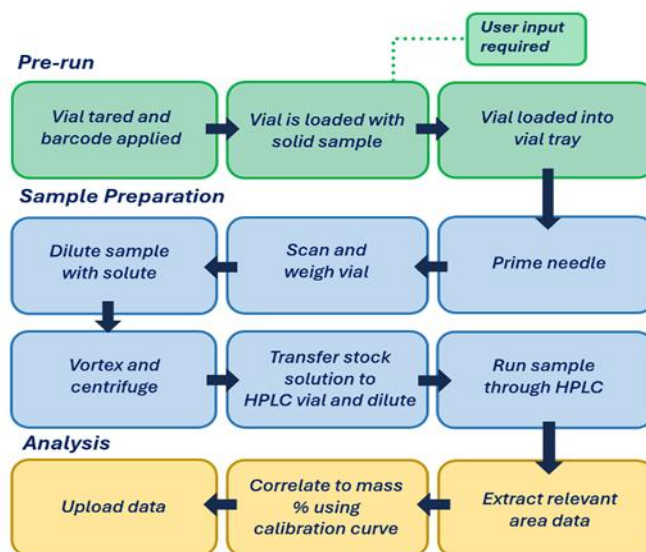


Figure X

manually load the desired compound into the vial, place it into the vial tray, and run the Python script to create the calibration curve.

The details of the Python code are as follows; italicized items indicate variables that can be varied in the python code:

1. The robot picks up a centrifuge vial from the specified position and brings it to the webcam to log the barcode.
2. The pre-tared vial with powder is transferred to the balance and weighed. The powder weight is logged in a JSON file.
3. To begin the dissolution process, the vial is returned to the vial tray and dissolved by a set amount of backing solvent specified by the *solvent_volume*. The vial is weighed again to determine the amount of solvent injected.
4. For dissolution, the vial is transferred to the vortexer and vortexed for an amount of time specified by *vortex_time*.
5. After vortexing, the robotic arm transfers the vial to be centrifuged at a specified speed specified by *speed*.
6. Before solution transfer, an HPLC vial is weighed and its mass is logged.
7. A small amount of sample, specified by *volume_ul*, is transferred from the centrifuge vial to the HPLC vial. The vial is weighed again and the mass of the transferred solute is determined.
8. To prevent HPLC saturation, the concentrated solution is diluted again by backing solvent specified by *dilution_volume*. The vial is weighed again to determine the added solution weight.
9. The robot moves the HPLC vial into the autosampler, and PyautoGUI starts the run.

- Peak area data and compound name is extracted from a .csv file output through the Agilent Data Analysis software.

For calibration curve generation, the peak area data for each HPLC run is plotted against the compound concentration, and the gradient is stored in a JSON file for later quantification. For quantification, the peak area data of each identified compound and concentration is calculated from the gradient of the corresponding calibration curve. The data processing procedure is described in detail in 2.2.5. For both cases, the backing solvent used to dissolve and dilute the samples was methanol. The HPLC system used a reverse-phase column with InfinityLab Poroshell 120 HPH (high PH) C18 stationary phase. The autosampler was programmed to sample a 1 μ L aliquot and quantification was done at detection wavelengths of 210 nm and 254nm.

3 Results and Discussion

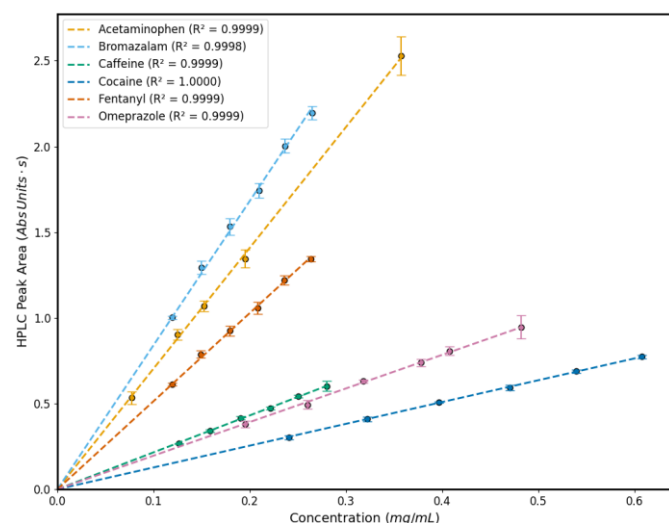


Figure X

3.1 Calibration Curve Workflow

Workflow validation was done by generating calibration curves using an HPLC system. The linearity of the calibration curves for eleven compounds (two over-the-counter medications and nine illicit compounds) was evaluated.

The HPLC method was optimized for separation of the eleven compounds of interest: Acetaminophen, Bromazepam, Caffeine, Cocaine, Cocaine HCl, Fentanyl, Ketamine, MDA (3,4-Methylenedioxyamphetamine), MDMA (3,4-Methylenedioxy methamphetamine), Methamphetamine, Omeprazole, and Parafluorofentanyl. The quantity and purity of the samples

were pre-assessed by H-NMR (proton nuclear magnetic resonance) spectroscopy.

Figure X below presents six calibration curves, based on their respective HPLC peak areas. The curves include trendlines and 95% confidence intervals, generated by plotting peak area data against sample concentrations. The compounds are divided into two graphs according to their absorption wavelength.

Linearity of the curves in Figure X was tested using regression analysis, and all curves gave a correlation coefficient R^2 of over 0.9997. The high correlation coefficient demonstrates that the preparation of the compounds is sufficiently precise for analytical purposes.

3.3 Quantification Examples

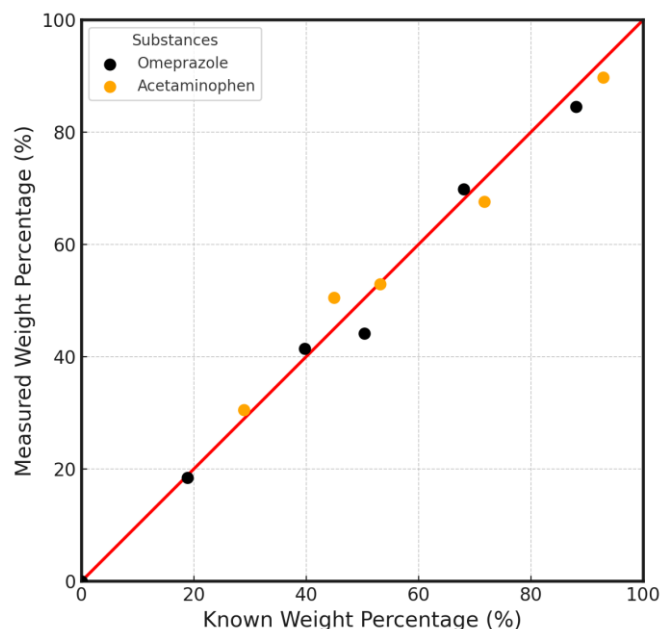


Figure X

The validity of our calibration curves was tested using Omeprazole and Acetaminophen. These compounds were mixed with known quantities of potato starch (which is relatively insoluble in methanol) to prepare mixtures with varying, known, mass percentages.

Figure X illustrates the quantification results for 14 test runs, including seven for Omeprazole and seven for Acetaminophen. The plotted line represents the theoretical outcome assuming perfect precision, where the experimental results match the theoretical values. The average error was calculated as 2.59% for Omeprazole and 2.41% for Acetaminophen. These results indicate that the system performs effectively in determining the mass percentages of unknown compounds.

3.4 Discussion

As demonstrated in the previous section, the robotic system can generate calibration curves from solid powders and subsequently quantify unknown samples autonomously, applicable to purposes like drug checking. The total run time of quantification for our current HPLC method is less than 15 minutes for the first sample, accounting for sample preparation and HPLC runtime. Subsequent runs 2-15 take approximately 9 minutes per sample for the per sample as the sample prep for the following sample is completed during the previous HPLC run. The run time for a calibration curve with each dilution run in triplicate is approximately 3 hours. The accuracy of the system is sufficiently high for laboratory and quantification purposes. The system can be especially useful when specialized labour is not readily available, as this versatile system requires minimal human input to quickly generate data.

In terms of performance, the system demonstrated remarkably reliable accuracy across a wide range of concentrations. During testing using a large variety of compounds, the robotic platform consistently generated calibration curves with a correlation coefficient over 0.9997 within three hours. The average standard deviation, expressed as a percentage of the total peak area for each triplicate run across six calibration curves, was 1.80%. This indicates minimal variability and reflects the reliability of the HPLC instrumentation. Similarly, the system can perform parallel quantifications of unknown samples accurately, with an average error of 2.49%, within 15 minutes per sample. For the quantification runs the mass of the known sample varied between 5 and 15 mg. The readability of the balance used in the system was 0.1 mg with a repeatability of 0.2 giving an error in the weight percent of 0.5 to 4%. Therefore, the combined uncertainty of the HPLC and balance are between 2 and 5.5%. When accounting for the combined uncertainties from the balance and the HPLC instrumentation, the system demonstrates excellent accuracy with only a minority portion of the error likely to originate in the automated sample prep and dilution.

While the system demonstrates promising results, several limitations remain, particularly with the software integration. Future work should focus on enhancing software compatibility with OpenLab CDS or exploring ways to directly interface with the HPLC backend. These efforts would further streamline data processing and reduce potential errors due to PyAutoGUI.

The functions of the system extend beyond the processing of solid powders, it can also be used to perform automated calibration curves for solutions. In this case, it is necessary to know the mass of the samples and solvents. It is also possible to generate calibration curves from impure powders given that the weight percentage is known. However, the system is not optimized for handling solid samples that do not readily dissolve in the backing solvent. Undissolved samples result in unreliable quantification and inaccurate calibration curves. Dissolution failure can be addressed by selecting a more suitable backing solvent. In such cases, the sample may remain cloudy or

precipitate at the bottom of the vials after vortexing, which can be observed visually. While our currently lacks such capabilities, the modular design of our system allows for the easy introduction a machine vision turbidity sensor like HeinSight, developed by El-khawaldeh, to automatically determine the level of compound dissolution.⁴²

4.0 Conclusions

In this work, we have introduced an end-to-end robotic system for analytical chemistry that integrates custom modules and commercial components to automate sample preparation and data analysis. Designed to replace manual input, our system, made to generate calibration curves and quantify unknown compounds, improves efficiency and reliability for some of the most common procedures in analytical chemistry.

Our validation experiments demonstrated remarkably high linearity in calibration curves across multiple compounds. Similarly, in quantification tests using multiple compounds, and only minor deviations between theoretical and measured were found. The precise validation results demonstrate the system's precision and reliability.

Apart from minimizing human labour, our work also features highly customizable, modular, and open-source preparatory equipment. Each component is built to function independently as a standalone unit and can be easily integrated into various workflows. For instance, the centrifuge or vortexer can be run as low-cost, standalone units by utilizing the code from their respective Python classes. The low cost and openly available modules enable researchers and labs with limited resources to access specialized equipment.

Looking ahead, we plan to expand our library to include a wider range of analytes to allow quantification of more compounds. Additionally, since it is possible for multiple compounds to absorb at similar retention times, we aim to extract providing compound-specific data by integrating a mass spectrometer (MS) to complement our HPLC. Furthermore, we aim to create a program that directly interfaces with the back end of Agilent OpenLab CDS software to eliminate the software failures due to PyAutoGUI limitations. We believe that by implementing these improvements into our system, our platform will be able to further position itself as a powerful analytical chemistry tool, opening the door to workflows that are more effective, reproducible, and democratized.

Author contributions

We strongly encourage authors to include author contributions and recommend using [CRediT](#) for standardised contribution descriptions. Please refer to our general [author guidelines](#) for more information about authorship.

Conflicts of interest

There are no conflicts to declare.

Data availability

Data used in this article, including code, 3D models, and HPLC chromatograph data used in this article are available at [Hein Group GitLab](#).

Acknowledgement

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